

The Base-Rate Trap — Gold Example

Worked for ClearList · hand out after the debrief

The one idea to leave with: at a low base rate, a high-TPR judge still produces a **low PPV** and a stream of **silent misses**. Accuracy on the judge says almost nothing about what the deployed system catches. **Zero flags is not zero failures**.

Step 1 · The matrix (50,000 alerts, 2% base rate)

Base rate splits the columns: 1,000 truly suspicious, 49,000 not. TPR 90% splits the first column; FPR 5% the second.

	TRULY SUSPICIOUS (1,000)	NOT SUSPICIOUS (49,000)	ROW TOTAL
Judge: ESCALATE	TP = 900	FP = 2,450	3,350
Judge: CLEAR	FN = 100	TN = 46,550	46,650
Column total	1,000	49,000	50,000

Step 2 · The numbers that matter

PPV = $900 / (900 + 2,450) = 900 / 3,350 = \approx 26.9\%$ → nearly 3 of every 4 escalations are false alarms.

FNR = $100 / 1,000 = 10\%$ (= 1 - TPR)

Truly-suspicious alerts auto-cleared each month (FN) = **100** → silently closed, never reviewed.

NPV looks reassuring at $46,550 / 46,650 \approx 99.8\%$ – which is exactly how the misses stay hidden.

A “90% accurate” judge escalates a pile that is only **27% real**, and quietly clears **100 suspicious cases a month**. Both problems come straight from the 2% base rate, not from a bad judge.

Step 3 · Effective sensitivity

Effective sensitivity = $0.90 \times 0.70 = 0.63$ → the system catches **63%** of truly suspicious activity, **misses 37%**.

False alarms flooding the queue = FP = **2,450 / month** (vs. 900 real).

Floods AND misses. The queue is 73% noise, so the 70% reviewer hit rate is optimistic — alert fatigue pushes it *down*, which lowers effective sensitivity further. A worse PPV makes the human worse too. The two failures compound.

Step 4 · “Zero flags last month”

No. The auto-cleared pile is never sampled, so “zero confirmed misses” only means **nobody looked**. At a 2% base rate the ≈100 monthly misses produce **no observable flags at all** — and with zero observed flags you can't even estimate the miss rate. (Prevalence corrections like Rogan–Gladen fix *measurement*, not *detection*: they won't tell you *which* 100 you missed.)

WHAT TO PUT IN THE MRM ARTIFACT

Not “judge accuracy 92%.” Report the **base rate**, the **PPV and FNR at that base rate**, the **effective sensitivity** (judge × reviewer), and a **random-audit stream over the auto-cleared pile** so the miss rate is observable at all.

Illustrative for the exercise only. The numbers are ClearList's stated figures; real judge performance must be measured on a labeled gold set, not assumed. Nothing here is legal or regulatory advice.